

WatchSchedule – Fairness Engine Functional Specification

Overview

The Fairness Engine is the core scheduling system within WatchSchedule.

Its purpose is to generate fair, defensible watch schedules while maintaining continuity through leave periods, crew changes and charter operations.

Unlike traditional rota systems, WatchSchedule does not evaluate fairness only within a single schedule.

The system continuously tracks fairness over time and uses historical scheduling data to influence future assignments.

Core Concepts

Crew Fairness Score™

A fairness score calculated for each crew member.

Range: 0–100

Purpose: Measure how fairly duties have been distributed to an individual over time.

Factors:

- Total duties completed
- Weekend duties completed
- Weighted duties completed
- Historical fairness
- Fairness debt
- Recent duty burden

Displayed: Per crew member

Updated: After every published schedule

Schedule Fairness Score™

An overall score representing the fairness of the proposed schedule.

Range: 0–100

Purpose: Provide confidence that the schedule is balanced before publication.

Factors:

- Duty distribution
- Weekend balance
- Consecutive duty compliance
- Fairness debt impact
- Rotation continuity

Displayed: Schedule level

Updated: During schedule generation

Duty Weighting System™

Not all duties carry the same burden.

The system supports configurable duty weightings.

Default values:

Weekday Duty: 1.0

Friday: 1.25

Saturday: 1.5

Sunday: 1.5

Public Holiday: 2.0

Christmas Day: 3.0

New Year's Day: 3.0

Weightings must be configurable per vessel.

Historical Fairness Score™

Measures fairness over time.

Purpose:

Prevent fairness being evaluated only within the current schedule.

Tracks:

- Previous schedules
- Previous seasons
- Historical weekend burden
- Historical duty burden

Updated after every schedule publication.

Fairness Debt™

Tracks additional burden carried by crew.

Purpose:

Prevent crew members being penalised for covering additional duties.

Examples:

- Covering leave
- Charter disruption
- Crew shortages
- Relief periods

Fairness debt increases when additional burden is carried.

Future schedules attempt to reduce fairness debt automatically.

Fairness debt can never be manually erased.

Only future duty allocation can reduce it.

Most Due To Serve™

Calculated value identifying who is most due for the next duty.

Factors:

- Crew Fairness Score
- Historical Fairness Score
- Fairness Debt
- Recent duty allocation
- Weekend burden

Displayed:

"Most Due"

Example:

Tom Smith

Fairness Debt: +3

Most Due To Serve

Crew Fairness Loop™

Every published schedule updates:

- Crew Fairness Score™
- Historical Fairness Score™
- Fairness Debt™

Purpose:

Create a self-correcting fairness system.

Every schedule influences future schedules.

Fairness is maintained across seasons rather than individual schedules.

Scheduling Rules

Consecutive Duties

Default:

Not permitted.

Example:

Monday Duty Tuesday Duty

Rejected.

Consecutive Weekend Mode

Optional.

Allows:

Saturday + Sunday

for same crew member.

Configurable.

Extended Weekend Mode

Optional.

Allows:

Friday + Saturday + Sunday

for same crew member.

Configurable.

Weekend Protection

Default:

A crew member assigned weekend duty should not be assigned adjacent Friday or Monday duties.

Examples:

Friday + Saturday Rejected

Sunday + Monday Rejected

Unless Extended Weekend Mode enabled.

Leave Neutral Scheduling

Crew taking leave must not be penalised.

The system must not allocate additional future duties solely because leave was taken.

Leave Coverage Fairness

Crew covering leave accumulate Fairness Debt credit.

Future schedules should restore balance automatically.

Charter Mode

Allows temporary suspension of normal scheduling.

When Charter Mode ends:

The system resumes from the correct continuity point.

The system recalculates fairness impact.

The system preserves historical fairness.

Relief Crew Handling

Relief personnel can be inserted into schedules.

Relief duties:

- Count toward schedule coverage
 - Do not permanently alter fairness calculations
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Rotation Continuity™

Purpose:

Maintain scheduling sequence despite operational disruption.

Events supported:

- Leave
- Charter periods
- Relief crew
- Joiners
- Leavers

System should:

Identify optimal restart point.

Minimise fairness impact.

Maintain scheduling continuity.

Generation Priorities

The scheduling engine should optimise in the following order:

Priority 1: Coverage achieved

Priority 2: Scheduling rules satisfied

Priority 3: Rotation continuity maintained

Priority 4: Fairness Debt reduced

Priority 5: Historical fairness improved

Priority 6: Schedule Fairness Score maximised

Future AI Layer

AI should never replace scheduling logic.

AI responsibilities:

- Explain assignments
- Explain fairness calculations
- Explain schedule decisions
- Recommend improvements

Examples:

Why was Tom assigned?

Because:

- Most Due To Serve
- Fairness Debt +3
- Weekend burden below average
- Continuity preserved

AI remains advisory.

Scheduling rules remain deterministic.